

Populations genetics

Biologia della Conservazione 25/26

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- ▶ Genetics (and genomic) have an important role in conservation biology and towards the protection of biodiversity
- ▶ Darwin was the first to consider the importance of genetics and evolution in the persistence of natural populations. He postulated that because of the reduced population size of deer in British natural reserves they may experience loss of vigor (Darwin, 1896)
- ▶ The modern concern for genetics in conservation began in the 1970s when Frankel began to raise the alarm about the loss of primitive crop varieties and their replacement by genetically uniform cultivars (Frankel, 1974; Frankel, Bennett, Food and Agriculture Organization of the United Nations, & International Biological Programme, 1970)

► Most articles dealing with conservation and genetics fit into one of the five general categories (Allendorf et al., 2022):

1. Management and reintroduction of captive populations, and the restoration of biological communities
2. Description and identification of individuals, genetic population structure, kin relationships, and taxonomic relationships
3. Detection and prediction of the effects of habitat loss, fragmentation, isolation, and genetic rescue
4. Detection and prediction of the effects of hybridization and introgression
5. Understanding the relationships between fitness of individuals or local adaptation of populations and environmental factors

- ▶ As in other sciences, conservation genetics has benefited from the use of model organisms for experiments as well as modelistic and numerical approaches
- ▶ How much gene flow is required to prevent the inbreeding effects of small population size?
 - ▶ We can use species with short generation time and easy to control (*Drosophila*, peas, *Arabidopsis thaliana*) to make empirical experiments and answer the above question

- ▶ Describing individuals from a molecular genetic standpoint can also help to better understand the basic biology of species and populations (Allendorf et al., 2022)
 - ▶ Total population size can be estimated from the genotypes in populations that are difficult to sample with the classic methodologies (Luikart, Ryman, Tallmon, Schwartz, & Allendorf, 2010)
- ▶ Genetic analysis can help detecting cryptic effects of climate change on the distribution of species (Allendorf et al., 2022)
 - ▶ Gurgel, Camacho, Minne, Wernberg, & Coleman (2020) documented the reduction in genetic diversity after a heatwave despite the sea weed population had recovered

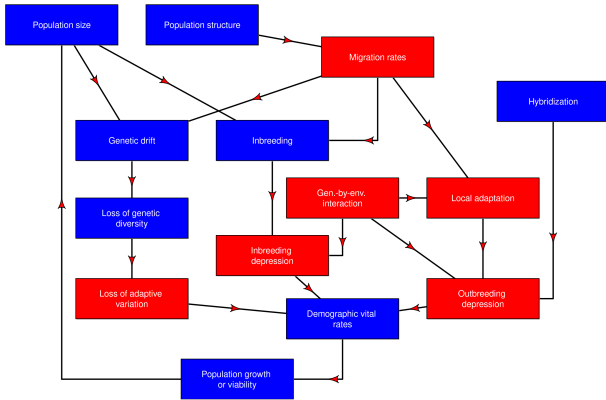


Figure 1: Genetic (blue) and genomic (blue) issues that can be addressed in a population

- ▶ In 1990 IUCN identified genetic diversity as one of the three most important element of conservation biology (Genetic diversity, Species diversity, Ecosystem diversity)
- ▶ **Phylogenetic diversity:** A monotypic genus receives higher priority (USA Fish and Wildlife Services, 1983). The Tuatara case from New Zealand is paradigmatic (Daugherty, Cree, Hay, & Thompson, 1990; Gemmell et al., 2020)
- ▶ Vane-Wright, Humphries, & Williams (1991) developed a method to assign a conservation value to distinct phylogenetic lineages. This value is inversely proportional to the number of branching points to other extant lineages

- ▶ **Species or ecosystems:** Conservation requires a balanced approach based on habitat protection that also takes into account the natural history and viability of individual species. This is particularly challenging for wide-ranging species (Allendorf et al., 2022)
- ▶ **Populations or species:** Genetically distinct populations should probably receive more attention than species from a conservation perspective
- ▶ The functional units of an ecosystem are populations

- ▶ Experiments conducted in the field of community genetics hint towards a correlation between individual alleles within species and community composition and diversity (Crutsinger, 2016)
- ▶ Genetic variation of a Tasmanian blue gum tree affects the distribution of insects, birds, and marsupials (Whitham et al., 2008)
- ▶ Conservation genetics ultimately looks for and tries to maintain the **allelic diversity** within a population and prevent the fixation of alleles

Term	Definition
Locus	The position on the chromosome where a specific gene is found, and/or the genetic segment, even a single nucleotide base, of interest.
Allele	Variant of a specific locus on homologous chromosomes (containing the same type of genetic information).
Genotype	Combination of alleles at one or more loci (genes) in an individual. For example, the genotype of a diploid individual characterized for the bi-allelic gene A might be A1A1, A1A2, or A2A2.

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Phenotype	The obvious and measurable manifestation of a specific genotype.
Genome	The set of nucleotide sequences, coding and not, generally organized into chromosomes, which characterize a species or an individual.
Homozygote	Individual with two identical copies of a specific allele registered at the locus (gene) of interest.
Heterozygote	Individual with two different alleles characterized at the locus (gene) of interest.
Allele frequency	The frequency of a specific allele in a population. The term gene frequency is often used to indicate the same concept.

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Polymorphic	The presence in a species or population of 2 or more alleles at a certain gene. For example, gene A in a population might have 4 alleles: A1, A2, A3, and A4. A locus is considered polymorphic if the frequency of its most frequent allele is less than 0.99 or 0.95.
Monomorphic	A population or species is monomorphic for a gene when all individuals have the same allele. In other words, the frequency of that allele in the population is 1.
Average heterozygosity	Average number of heterozygous loci in an individual or population.

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Co-dominance	Situation where all genotypes can be identified by their phenotypes. In the case of gene A with two codominant alleles, the A1A1, A1A2, and A2A2 genotypes all produce different and distinguishable phenotypes.
Allelic diversity	Mean number of alleles per locus.
Genetic distance	A measure of distance obtained by comparing 2 or more nucleotide sequences or by comparing the allele and genotype frequencies of two or more individuals or populations.

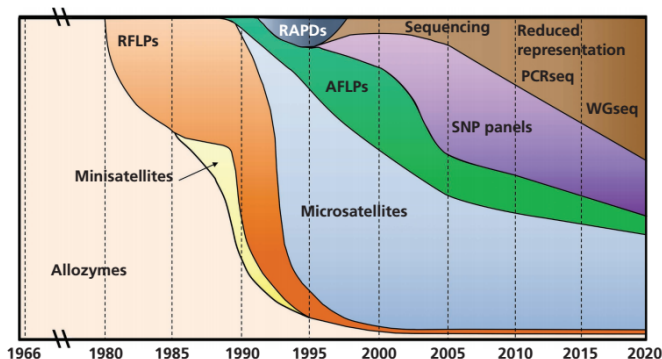


Figure 4.1 Relative popularity of molecular markers in conservation genetics over time. RFLP, restriction fragment length polymorphism; AFLP, amplified fragment length polymorphism; RAPD, randomly amplified polymorphic DNA; SNP, single nucleotide polymorphism. SNP panels include array-based or PCR-based genotyping (by hybridization or Fluidigm microfluidics). Sequencing is subdivided into three approaches: PCRseq (Sanger sequencing of single or few loci), reduced representation sequencing (mainly RADseq and also targeted DNA capture and amplicon sequencing), and whole genome resequencing (WGseq). Modified from [Schlötterer \(2004\)](#).

Figure 2: Figure reproduced from Allendorf et al. (2022)

- ▶ Polymorphism

$P = \text{number of polymorphic loci} / \text{total number of loci}$

- ▶ Allele frequency (for diploid individuals)

$$p = \frac{2n_{Hom} + n_{Het}}{2n}$$

- ▶ Average heterozygosity $H = \sum H_i / N$ where H_i is the average heterozygosity at locus i , and N is the total number of loci

- ▶ Describing the genetic variation using mathematical tool is a good and necessary exercise, but it is not enough!
- ▶ We need to develop models that could tell us about how various evolutionary factors (natural selection, mutations, population size, geographic fragmentation, drift) affect the persistence of populations and species
- ▶ Models are essential for understanding the genetics of natural populations in a variety of ways: *i)* models make us define the parameters that need to be considered; *ii)* models allow us to test hypotheses; *iii)* models allow us to generalize results; *iv)* models allow us to predict how a system will operate in the future (Allendorf et al., 2022, p. 95).

The simpler model: Hardy-Weinberg principle



Figure 3: Gregory Harold Hardy, figure from the Wiki page

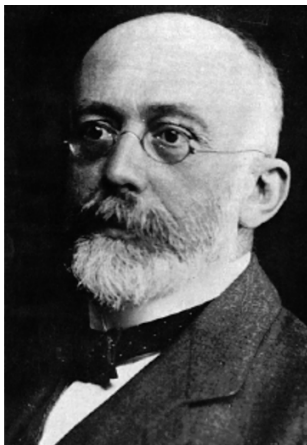


Figure 4: Wilhelm Weinberg, figure from <http://www.esp.org/people/>

Assumption	Significance
Random mating	Individuals show no significant preference or dislike for other individuals with respect to specific traits or loci.
No mutation	Mutation is one of the few forces creating variability. We assume that the information is transmitted from generation to generation without the influence of mutation.
Infinite population size	No population in nature is actually infinite (although populations of some species may as well be)! This assumption is fundamental to construct a null model and compare the effect of small population size.

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No natural
selection

We will assume that all alleles and genotypes have the same fitness and that there is no differential reproduction or survival. This assumption is very unlikely to be true in nature, especially because selection may be reduced for the loci under analysis, but still present at all other loci in an individual.

No immigration

We will first consider a situation where only one population of a certain species is present. This means that we will not consider gene flow and exchange of genetic material from and to other populations.

! Question

What is the main consequence of the assumptions we have just described?

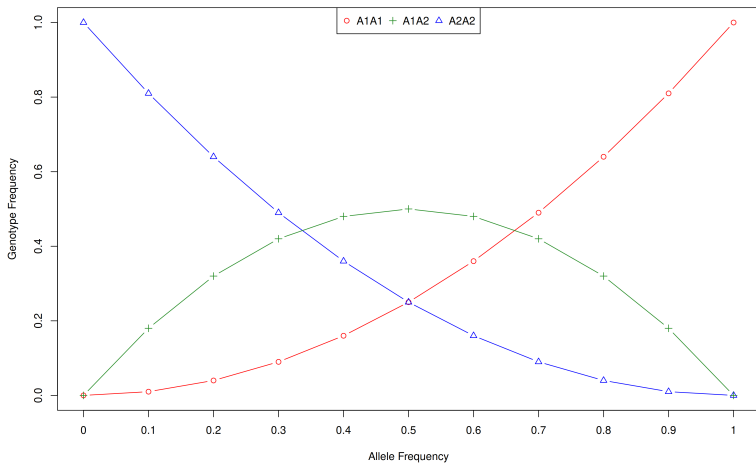


Figure 5: Organization of genetic variability in a Hardy-Weinberg equilibrium population typed at a biallelic locus.

Analyzing the Hardy-Weinberg model

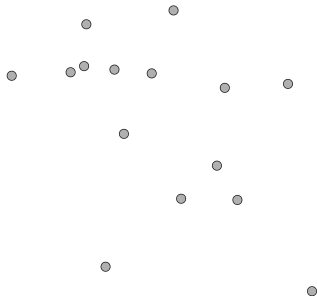


Figure 6: Diploid population with $n=15$

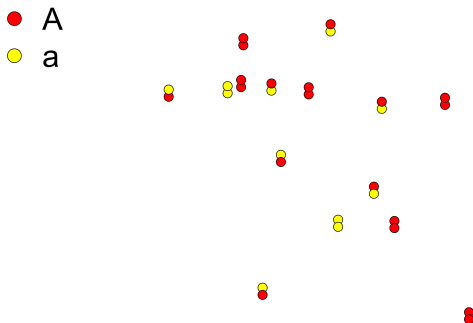


Figure 7: Two alleles per locus (A and a)

n_{AA}	n_{Aa}	n_{aa}	n
6	7	2	15

$$p = f(A) = \frac{2n_{AA} + n_{Aa}}{2n} = \frac{12 + 7}{30} \approx 0.633$$

$$q = f(a) = \frac{2n_{aa} + n_{Aa}}{2n} = \frac{4 + 7}{30} \approx 0.367$$

$$p + q = 1$$

- ▶ Because of our assumption of random mating the assortment of alleles in the zygotes can be estimated from the allele frequencies using a binomial expansion:

$$(p + q)^2 = p^2 + 2pq + q^2$$

- ▶ If we want to add another allele (A_1, A_2, A_3)...

$$(p + q + r)^2 = p^2 + 2pq + q^2 + 2pr + 2qr + r^2$$

- Random mating and same fitness mean the frequencies of male and female gametes correspond to the allele frequencies and zygotes are formed randomly choosing gametes from the gene pool

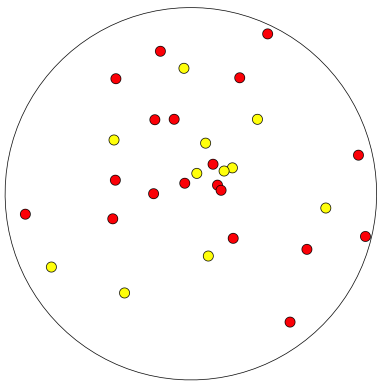


Figure 8: Gene pool

- ▶ The probability of two independent events, such as the union of two gametes at random in a zygote, is calculated as the product of the probability of the single events

```
(dice <- c(1,2,3,4,5,6))
```

```
[1] 1 2 3 4 5 6
```

```
sample(dice, size = 1)
```

```
[1] 3
```

```
sample(dice, size = 1)
```

```
[1] 2
```

```
sim_dice <- sample(dice, size = 100000, replace = T)
table(sim_dice)/100000
```

```
sim_dice
      1      2      3      4      5      6
0.16958 0.16426 0.16527 0.16674 0.16689 0.16726
```

- ▶ The probability of casting a dice and obtaining one of the six numbers is $\frac{1}{6} \approx 0.17$
- ▶ The probability of casting the dice twice and obtaining the same number is $\frac{1}{6} * \frac{1}{6} = \frac{1}{36} \approx 0.028$

- We want to recreate a sample of 15 individuals starting from the gamete pool of our original population with $n = 15$ and $p \approx 0.633$ and $q \approx 0.367$



```
p <- 0.633
q <- 0.367
sample(c("A", "a"), size = 1, prob = c(p, q))
```

- Every time we sample an allele, we add a color. Remember A = red and a = yellow. An animation of the process is available [here](#)





n_{AA}	n_{Aa}	n_{aa}	n
7	7	1	15

- The genotype and allele frequencies of the new generation are different from the previous one. We repeat the experiment one more time

```
p <- 0.633
q <- 0.367
newSample <- sample(c("A", "a"), size = 30,
                    prob = c(p, q),
                    replace = T)
table(newSample)
```

```
newSample
  a  A
10 20
```


- Once again, the genotype and allele frequencies of the new generation are different from the previous one, and it will likely be different most of the time we do this again

```
p <- 0.633
q <- 0.367
newSample <- sample(c("A", "a"), size = 30,
                    prob = c(p, q),
                    replace = T)
table(newSample)
```

```
newSample
  a  A
11 19
```

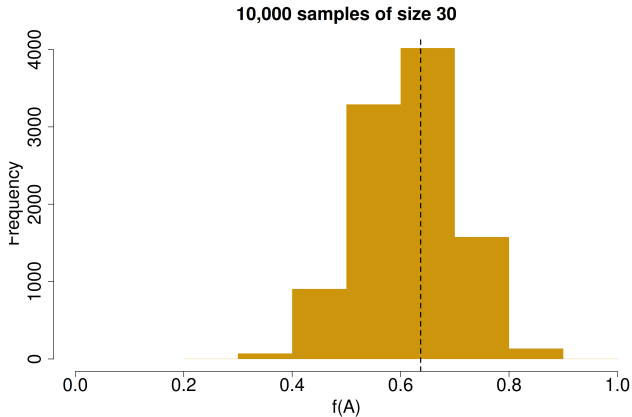


Figure 9: There is a large variability around the original value of $p = 0.633$

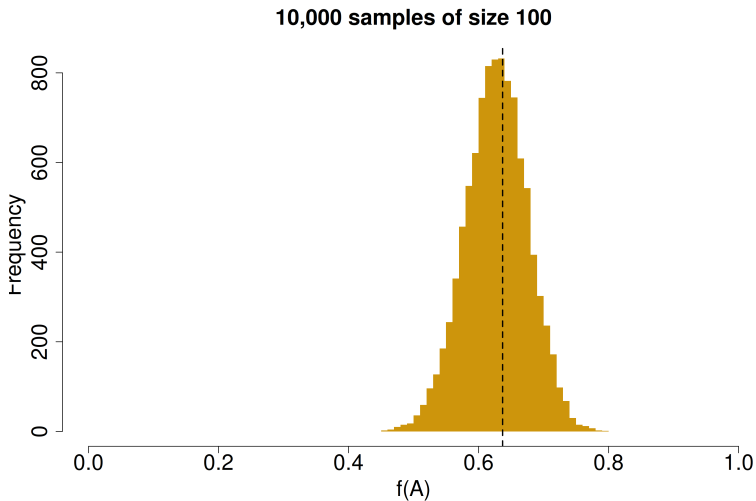


Figure 10: Smaller variability if we collect a larger sample

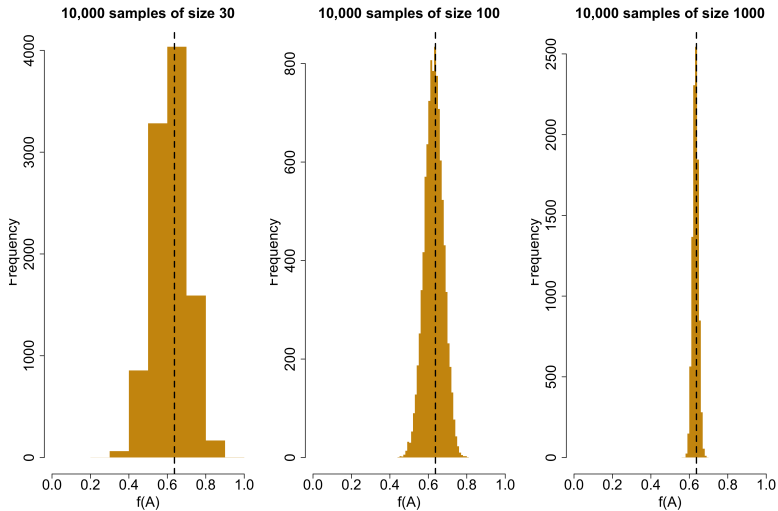


Figure 11: Sample size matters to prevent loss of genetic variability

Summary of the Hardy-Weinberg model

- ▶ The larger the sample dimension, the lower is the variation in allele frequencies with respect to the original value
- ▶ If all the assumptions of the HW model are met, we should expect no variation at all in allele frequency, generation after generation
- ▶ In natural population it is very hard to meet all the assumptions of the HW
- ▶ The HW model is a very effective and simple way to “see” the effects of various evolutionary forces operating at the population level
- ▶ With this in mind we can start investigate the relative role of the various evolutionary forces

- ▶ Genetic drift is the random change in allele frequencies from generation to generation because of sampling error. That is, the finite number of genes transmitted to progeny will be an imperfect sample of the allele frequencies in the previous generation (Allendorf et al., 2022, p. 114)
- ▶ Genetic drift is the primary cause bringing about allele frequency changes throughout the genome over time (Allendorf et al., 2022; Chen et al., 2018)
- ▶ With many samples ($n \gg 1$) the probability that their alleles all are A_1 or A_2 becomes $(\frac{1}{2})^{2n}$, and it will be less likely to lose one of the two alleles

- ▶ The probability that a population has k copies of allele A_1 and $2n - k$ copies of A_2 allele is $\binom{2n}{k} p^k q^{2n-k}$ where $\binom{2n}{k}$ is the binomial function $\frac{2n!}{k!(2n-k)!}$
- ▶ The variance around the mean allele frequency is the binomial sampling variance: $\sigma_p^2 = \frac{p_0 q_0}{2n}$
- ▶ Variances are higher in smaller than larger populations
- ▶ In small population the process of sampling happens every single generation, and its effect on allele frequency is cumulative

- ▶ Genetic drift will eventually cause all except one allele to go to fixation.
- ▶ The probability that a gamete does not contain allele A_1 (with frequency p in the parental population) is $1 - p$

$$P(\text{fixation } A_2) = (1 - p)^{2n}$$

! Problem

Diploid population with $n = 4$ with allele A_1 with a frequency $p = 0.25$ and allele A_2 with a frequency $q = 0.75$. What is the probability of losing allele A_1 in one generation of sampling to recreate a population of the same size?

 Solution

$$(1 - 0.25)^8 = 0.1001129$$

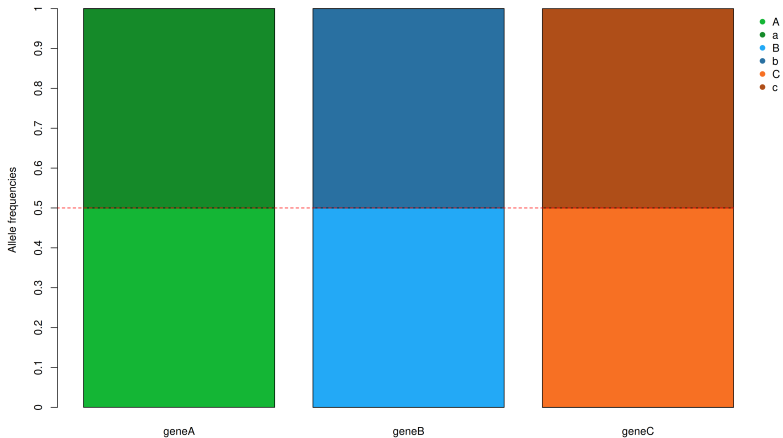


Figure 12: Three alleles in a population at $t = 0$

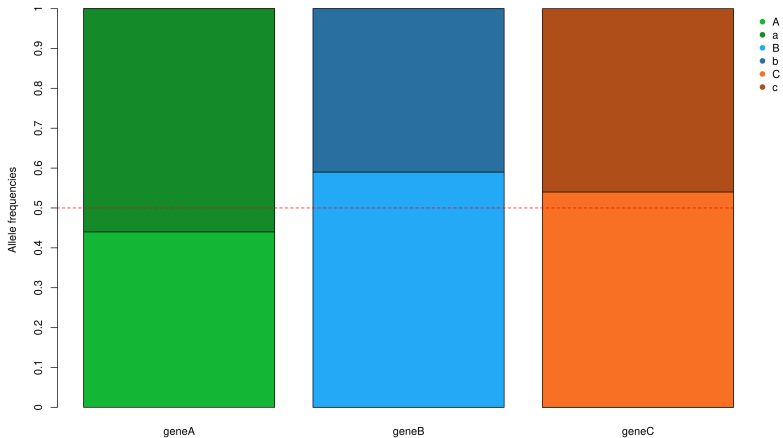


Figure 13: Three alleles in a population at $t = 1$

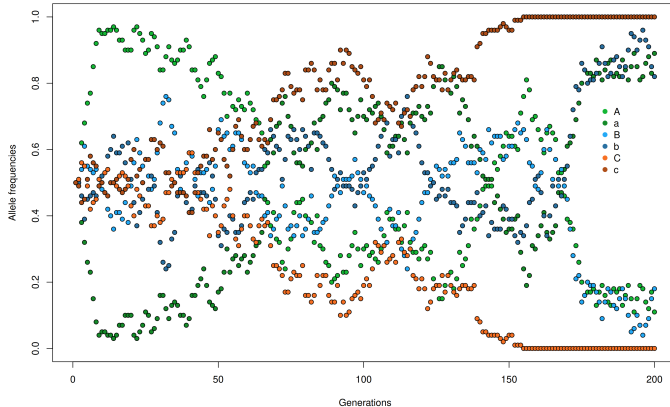


Figure 14: Allele frequencies variation after 200 generations while maintaining $N = 100$ (Russo, Colosimo, Fanelli, Franceschini, & Parisi, 2021)

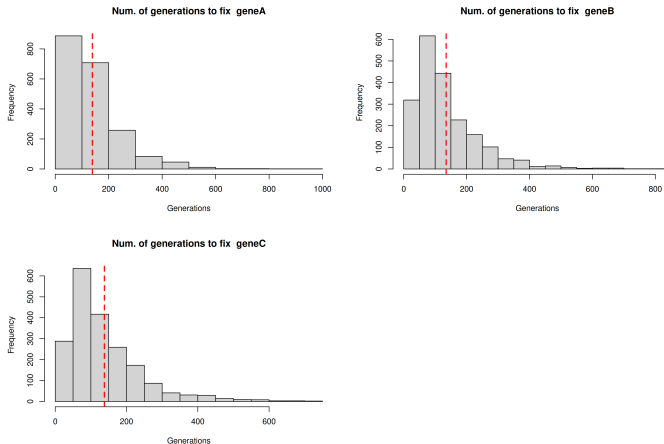


Figure 15: Simulating the average time of allele fixation in our example (Russo et al., 2021)

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